

## Experiment: 8: Steering Angle Variation in MIMO Beamforming Using MATLAB

### Objective – 1

```
clc;

clear;

close all;

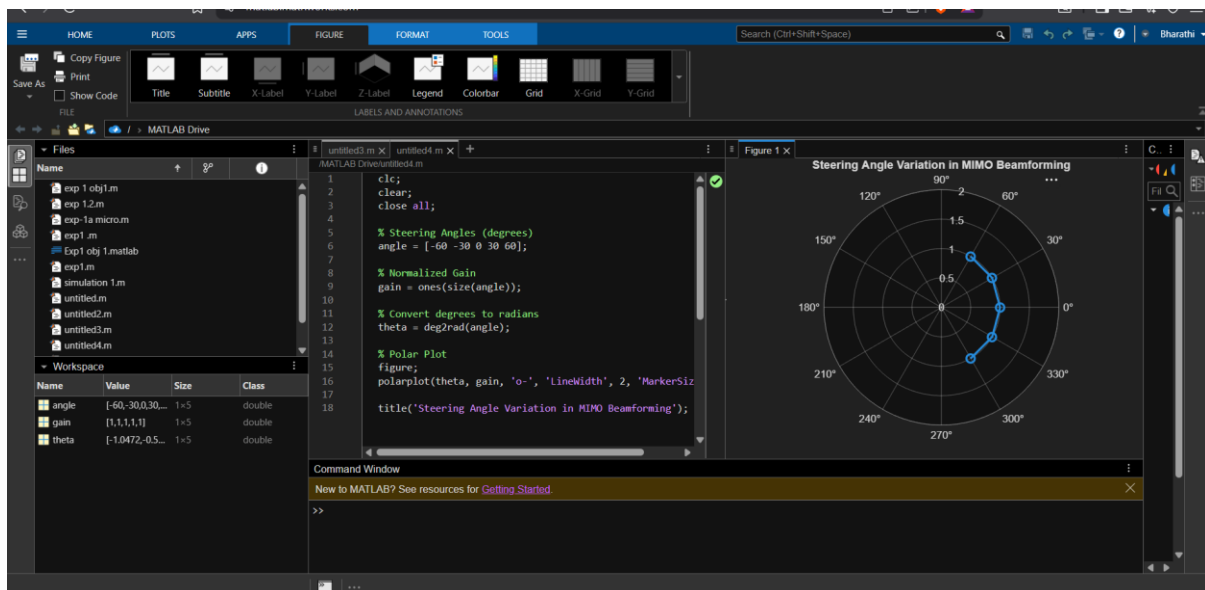
angle = [-60 -30 0 30 60]; % Steering Angles (degrees)

gain = ones(size(angle)); % Normalized Gain

figure

polarplot(deg2rad(angle), gain,'o-','LineWidth',2)

title('Steering Angle Variation in MIMO Beamforming')
```



### Objective – 2

```
clc;

clear;

close all;

angle = [-60 -30 0 30 60]; % Steering Angles (degrees)

gain = [9 12 15 12 9]; % Array Gain (dB)

figure
```

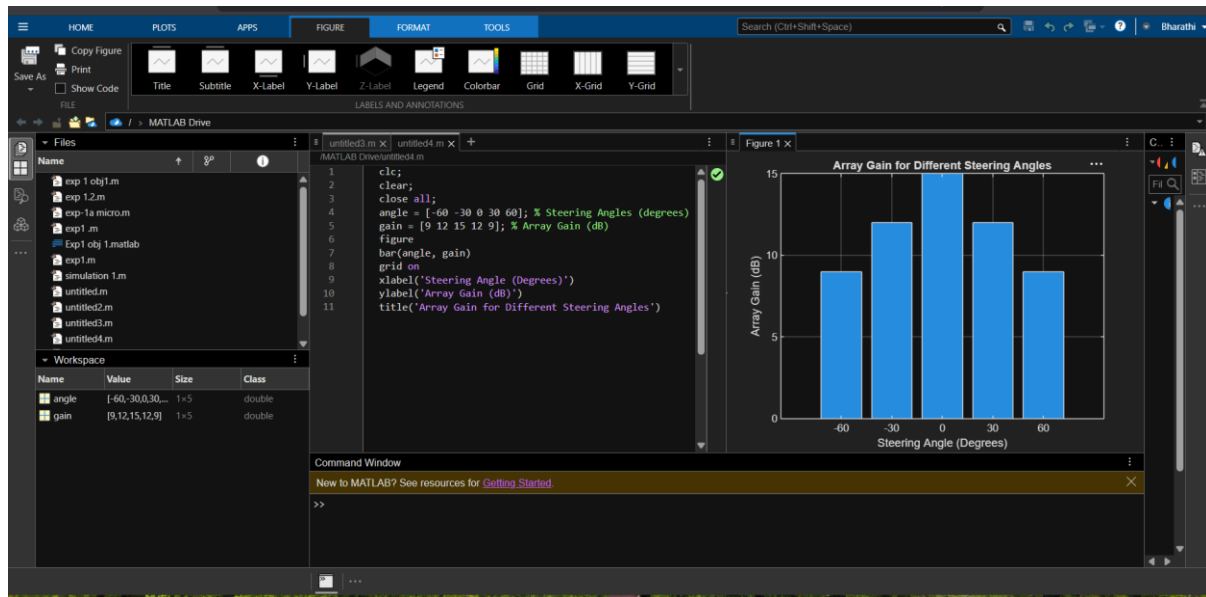
bar(angle, gain)

grid on

xlabel('Steering Angle (Degrees)')

ylabel('Array Gain (dB)')

title('Array Gain for Different Steering Angles')



## Objective – 3

clc;

clear;

close all;

desired = [-60 -30 0 30 60]; % Desired Steering Angle (degrees) actual = [-58 -32 1  
29 62]; % Actual Beam Direction (degrees)

error = abs(desired - actual); % Beam Direction Error

figure

plot(desired, error, 'o-', 'LineWidth', 2)

grid on

xlabel('Desired Steering Angle (Degrees)')

ylabel('Beam Direction Error (Degrees)')

title('Beam Direction Error for Different Steering Angles')

